# Java Collection-Based Mini Projects – Complete Startup Guide

This document provides detailed guidance to START each project using ONLY Java Collections. No database, no Spring, no UI — purely Core Java with List, Set, Map, and Queue. Each project includes purpose, entities, collections to use, operations, and menu flow.

## 1. Employee Management System

Purpose:
To manage employee records in memory using Java collections.

Entities:
Employee → id, name, age, salary, designation

Collections to Use:
- ArrayList to store employees
- HashMap for fast ID lookup

Core Operations:
- Add employee (ID must be unique)
- View all employees
- Search employee by ID
- Update employee details
- Delete employee

Menu Flow:
1. Add Employee
2. View All Employees
3. Search Employee by ID
4. Update Employee
5. Delete Employee
6. Exit

Validations:
- Prevent duplicate ID
- Handle invalid ID using exceptions

## 2. Student Result Processing System

Purpose:
To calculate and analyze student academic results.

Entities:
Student → rollNo, name, marks (3 subjects)

Collections to Use:
- ArrayList
- TreeMap for ranking

Core Operations:
- Add student marks
- Calculate total and average
- Assign grade
- Sort students by marks
- Display topper list

Menu Flow:
1. Add Student
2. Calculate Result
3. Display All Results
4. Show Toppers
5. Exit

## 3. Library Book Management System

Purpose:
To manage books and issue/return process.

Entities:
Book → bookId, title, author, status

Collections to Use:
- HashMap

Core Operations:
- Add new book
- Issue book
- Return book
- Search by title/author

Validations:
- Prevent issuing already issued book

# 4. Online Shopping Cart

Purpose:
To simulate a shopping cart system.

Entities:
Product → productId, name, price, quantity

Collections to Use:
- HashMap

Core Operations:
- Add product to cart
- Remove product
- Update quantity
- Calculate total bill

## 5. Contact Management System

Purpose:
To manage personal contacts.

Entities:
Contact → name, phone, email

Collections to Use:
- TreeMap

Core Operations:
- Add contact
- Search contact
- Delete contact
- Display contacts alphabetically

## 6. Banking Account Management System

Purpose:
To simulate basic banking operations.

Entities:
Account → accountNo, holderName, balance

Collections to Use:
- HashMap

Core Operations:
- Create account
- Deposit money
- Withdraw money
- Check balance

Validations:
- Minimum balance check

## 7. Movie Ticket Booking System

Purpose:
To manage movie seat bookings.

Entities:
Movie → movieName, availableSeats

Collections to Use:
- HashMap

Core Operations:
- View movies
- Book tickets
- Cancel booking

# 8. College Course Registration System

Purpose:
To manage course enrollment.

Entities:
Student, Course

Collections to Use:
- HashMap>

Core Operations:
- Enroll course
- Drop course
- View registered courses

## 9. Inventory Management System

Purpose:
To track product inventory.

Entities:
Item → itemId, name, stock

Collections to Use:
- HashMap

Core Operations:
- Add stock
- Reduce stock
- Low-stock alert

## 10. Voting System Simulation

Purpose:
To conduct fair voting.

Entities:
Voter → voterId
Candidate → name, votes

Collections to Use:
- HashSet for voted voters
- HashMap for votes

Core Operations:
- Cast vote
- Count votes
- Declare winner

## 11. Task / To-Do List Manager

Purpose:
To manage tasks by priority.

Entities:
Task → taskId, description, priority

Collections to Use:
- PriorityQueue

Core Operations:
- Add task
- View next task
- Complete task

## 12. Online Examination System

Purpose:
To conduct online exams.

Entities:
Question → question, options, answer

Collections to Use:
- ArrayList

Core Operations:
- Display questions
- Accept answers
- Calculate score

## 13. Attendance Management System

Purpose:
To track daily attendance.

Entities:
Student → rollNo, name

Collections to Use:
- HashMap>

Core Operations:
- Mark attendance
- Generate report

## 14. Hotel Room Booking System

Purpose:
To manage room bookings.

Entities:
Room → roomNo, type, status

Collections to Use:
- HashMap

Core Operations:
- Book room
- Cancel booking
- View availability

## 15. Customer Feedback & Rating System

Purpose:
To analyze customer feedback.

Entities:
Feedback → customerName, rating, comment

Collections to Use:
- ArrayList
- TreeMap>

Core Operations:
- Add feedback
- Calculate average rating
- Display best ratings